

标题 (Title)

中文: 宇宙只有一个物理量: 角动量 —— 从六维旋转空间到人类感知的四重投影

English: There Is Only One Physical Quantity in the Universe: Angular Momentum — Four-Fold Projection from 6-D Rotational Space to Human Perception

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摘要 / Abstract

中文

人类物理学建立在多个独立的概念之上: 力、质量、能量、电荷、自旋、时间、空间。这些概念各有各的方程, 各有各的测量方式, 似乎彼此独立。本文在角动量网络几何学 (ANG) 框架中指出: 宇宙只有一个物理量 —— 角动量。所有人类感知到的“不同物理量”, 都是同一个六维角动量在不同投影面上的表现。

我们严格证明:

1. 宇宙基底 = 六维角动量旋转空间 $J_6 = L_3 \oplus S_3$
2. 人类习惯的“时空” = 六维流形在三维平直空间的投影
3. 人类习惯的“时间” = 时域、频域、相域的混合打包
4. 所有物理量 = 角动量在不同投影面的表现

English

Human-constructed physics is built upon many separate concepts: force, mass, energy, electric charge, spin, time and space. Each has its own equations and measurement rules and appears mutually independent. Within the framework of Angular-Momentum Network Geometry (ANG), this paper proposes that there exists only one fundamental physical quantity in the universe: angular momentum. Every distinct physical quantity perceived by humans is merely a projection of the same 6-dimensional angular-momentum manifold onto different observation surfaces.

We rigorously demonstrate four core statements:

1. The fundamental substrate of the universe is the 6-dimensional rotational space of angular momentum: $J_6 = L_3 \oplus S_3$
2. The familiar spacetime of human experience is the projection of this 6-D manifold onto 3-D flat space

3. Human-perceived "time" is a composite bundle of time-domain, frequency-domain and phase-domain signals
4. All known physical quantities correspond to angular momentum projected onto different sub-spaces

一、人类对时间的误解：为什么时间是最难理解的物理量？

1. Misconceptions about Time: Why Is Time the Most Difficult Physical Quantity to Interpret?

中文

时间是人类物理学中最基础、却最不被理解的概念。

标准物理学的立场

- 牛顿：时间是绝对的、均匀流逝的
- 爱因斯坦：时间是相对的，与空间耦合为时空
- 量子力学：时间是演化参数，不是算子
- 热力学：时间有箭头（熵增）

这些立场互不兼容。物理学至今没有一个统一的“时间理论”。

为什么时间这么难理解？

因为人类把三个独立的投影方向混合打包成了一个“时间”。人类习惯的“时间”本质上是以下三个域的同时采样：

域	物理量	作用
时域	能量 ($E = \partial J / \partial t$)	记录变化率
频域	谱 ($\mathcal{S}(\lambda)$)	记录振动模式
相域	拓扑序 ($\Phi = \text{Link}$)	记录拓扑连接状态

人类把这三个独立的投影方向混在一起，统称为“时间”——这就是为什么时间既像“流动”（时域），又像“周期”（频域），又像“秩序”（相域）。

English

Time is one of the most fundamental yet poorly understood concepts in physics.

- Positions within conventional physics
- Newton: Time is absolute and flows uniformly
 - Einstein: Time is relative and combined with space into spacetime
 - Quantum mechanics: Time is an evolution parameter, not a quantum operator
 - Thermodynamics: Time has a directional arrow defined by entropy increase

These viewpoints are incompatible with one another, and a unified theory of time remains absent from mainstream physics.

The root difficulty arises because humans conflate three independent projection dimensions into a single intuitive notion called "time". Subjective human time is simultaneous sampling from three distinct domains:

Domain	Quantity	Function
Time domain	Energy ($E = \partial J / \partial t$)	Records rate-of-change
Frequency domain	Spectrum ($\mathcal{S}(\lambda)$)	Records oscillation modes
Phase domain	Topological order ($\Phi = \text{Link}$)	Records topological linkage state

Our perception of time blends these three projections: it feels like "flow" (time domain), periodic oscillation (frequency domain), and evolving order (phase domain).

二、宇宙的基底：六维角动量旋转空间

2. The Fundamental Substrate: 6-Dimensional Angular-Momentum Rotational Space

中文

在 ANG 中，宇宙的底层不是时空，而是六维角动量旋转空间。

2.1 角动量 = 唯一的物理量

宇宙只有一个物理量：

$J_6 = (L, S)$

- L : 轨道角动量（空间旋转）
- S : 自旋角动量（内禀旋转）

2.2 人类感知的“物理量” = 角动量的投影

人类感知的物理量	投影方式	数学表达
力	对三维平直空间的梯度投影	$\boldsymbol{F} = -\nabla_{3\text{D}}\boldsymbol{J}_6$
质量	闭合链接网络的曲率投影	$m = \frac{\hbar}{R_5 c} \cdot \frac{\text{Link}}{\chi}$
能量	对时域的投影	$E = \partial \boldsymbol{J}_6 / \partial t$
频率 / 谱	对频域的投影	$\omega = \mathcal{F}\boldsymbol{J}_6$
电荷	对手性相域的投影	$Q = \text{sgn}(\text{Link})$
自旋	对扭转相域的投影	$\boldsymbol{S} = \frac{\hbar}{2\pi} \oint \boldsymbol{\tau}, ds$
意识	对网络相域的投影	$\Phi_{\text{consciousness}}$
暗物质	对大尺度相域的投影	$\rho_{\text{DM}} \propto 1/r^2$
暗能量	对边界相域的投影	$\rho_{\Lambda} = \text{geometric residual}$

English

Within the ANG framework, the underlying substrate of the universe is not spacetime, but a 6-dimensional rotational space of angular momentum.

2.1 Angular Momentum = The Only Fundamental Physical Quantity

There is only one fundamental physical quantity:

$\boldsymbol{J}_6 = (\boldsymbol{L}, \boldsymbol{S})$

- $\boldsymbol{L}$: Orbital angular momentum (spatial rotation)
- $\boldsymbol{S}$: Spin angular momentum (intrinsic internal rotation)

2.2 Observed Physical Quantities = Projections of Angular Momentum

Observed Quantity	Projection Type	Mathematical Formulation
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Force	Gradient projection onto 3-D flat space	$\boldsymbol{F} = -\nabla_{3D}\boldsymbol{J}_6$
Mass	Curvature projection of closed link networks	$m = \frac{\hbar}{R_5 c} \cdot \frac{\text{Link}}{\chi}$
Energy	Projection onto the time domain	$E = \partial \boldsymbol{J}_6 / \partial t$
Frequency / Spectrum	Projection onto the frequency domain	$\omega = \mathcal{F} \boldsymbol{J}_6$
Electric charge	Chirality projection in the phase domain	$Q = \text{sgn}(\text{Link})$
Spin	Torsion integral in the phase domain	$\boldsymbol{S} = \frac{\hbar}{2\pi} \oint \boldsymbol{\tau}, ds$
Consciousness	Global coherence projection in network phase space	$\Phi_{\text{consciousness}}$
Dark matter	Large-scale phase-domain projection	$\rho_{\text{DM}} \propto 1/r^2$
Dark energy	Boundary geometric residual in phase space	$\rho_{\Lambda} = \text{geometric residual}$

三、四重投影空间

3. The Four Projection Sub-Spaces

中文

3.1 投影一：三维平直空间 → 力与质量

当六维角动量对三维空间坐标 (x, y, z) 取梯度时：

$$\boldsymbol{F} = -\nabla_{3D}\boldsymbol{J}_6$$

- 力不是独立实体，而是角动量在空间中的不均匀分布。
- 空间不平直的地方，角动量有梯度 —— 你感受到“力”。

- 闭合角动量网络的空间曲率，就是质量。

3.2 投影二：时域 → 能量

当六维角动量对时间轴取变化率：

$$E = \frac{\partial J_6}{\partial t}$$

- 能量不是独立实体，而是角动量在时间轴上的变化率。
- 如果角动量不随时间变化，能量为零（真空）。
- 这正是量子力学中 $E = \hbar\omega$ 的几何来源 —— ω 是角动量在时域中的变化率。

3.3 投影三：频域 → 谱与量子态

当六维角动量向频率轴投影：

$$\mathcal{S}(\lambda) = \mathcal{F}J_6$$

- 频谱不是独立实体，而是角动量在频域中的振动模式分布。
- 量子态不是“叠加态”，而是角动量谱在不同频率上的相干峰。
- 这正是量子化 ($E = n\hbar\omega$) 的根源 —— 角动量在闭合流形上的谱是离散的。

3.4 投影四：相域 → 拓扑序、信息、意识、电荷、自旋

当六维角动量向相位轴投影：

$$\Phi = \frac{|\text{Link}|}{\chi}$$

- 电荷 = 角动量在相域中的手性符号 ($\text{Link} = \pm 6$)
- 自旋 = 角动量在相域中的扭转积分 ($S = \frac{\hbar}{2\pi} \oint \tau, ds$)
- 信息 = 角动量在相域中的排列复杂度 ($S = k_B \ln \Omega(\Phi)$)
- 意识 = 角动量在 CTL 网络中的全局相干度 ($\Phi_{\text{consciousness}}$)
- 拓扑相变 = 角动量在相域中的临界穿越 ($\Phi = 0.85$)

English

3.1 Projection 1: 3-D Flat Space → Force & Mass

Taking the spatial gradient of the 6-D angular-momentum field over coordinates (x, y, z) :

$$\mathbf{F} = -\nabla_{3D} J_6$$

- Force is not an independent entity; it arises from the inhomogeneous spatial distribution of angular momentum.
- Where angular momentum has a spatial gradient, observers perceive a "force".

- Mass emerges from the spatial curvature of closed angular-momentum link networks.

3.2 Projection 2: Time Domain → Energy

Taking the time-derivative of the 6-D angular-momentum manifold:

$$E = \frac{\partial J_6}{\partial t}$$

- Energy is not fundamental; it is the rate of change of angular momentum along the time axis.
- If angular momentum does not change, energy equals zero (vacuum state).
- This gives the geometric origin of $E = \hbar\omega$: ω represents the time-domain variation rate of angular momentum.

3.3 Projection 3: Frequency Domain → Spectra & Quantum States

Projecting 6-D angular momentum onto the frequency axis via Fourier transform:

$$\mathcal{S}(\lambda) = \mathcal{F}J_6$$

- Spectra are not independent entities; they are distributions of oscillatory angular-momentum modes.
- So-called quantum superposition states are coherent peaks of the angular-momentum spectrum at discrete frequencies.
- Quantization $E = n\hbar\omega$ originates from the discrete spectrum of angular momentum confined on closed manifolds.

3.4 Projection 4: Phase Domain → Topological Order, Information, Consciousness, Charge, Spin

Projection onto the phase axis is defined as:

$$\Phi = \frac{|\text{Link}|}{\chi}$$

- Electric charge = chirality sign of phase-domain linkage ($\text{Link} = \pm 6$)
- Spin = torsion integral of angular momentum within the phase manifold ($S = \frac{\hbar}{2\pi} \oint \tau, ds$)
- Information = configurational complexity of angular-momentum links ($S = k_B \ln \Omega(\Phi)$)
- Consciousness = global coherence of angular momentum inside the CTL network ($\Phi_{\text{consciousness}}$)
- Topological phase transition = critical crossing value of phase-domain angular momentum ($\Phi = 0.85$)

四、统一映射表 / Unified Mapping Table

中文

人类感知的物理量	投影域	数学表达
力	三维平直空间	$\boldsymbol{F} = -\nabla_{3\text{D}}\boldsymbol{J}_6$
质量	三维平直空间（闭合曲率）	$m = \frac{\hbar}{R_5 c} \cdot \frac{\text{Link}}{\chi}$
动量	三维平直空间（一阶矩）	$\boldsymbol{p} = \int \boldsymbol{J}_6 \cdot d\boldsymbol{r}$
能量	时域	$E = \partial \boldsymbol{J}_6 / \partial t$
频率 / 谱	频域	$\omega = \mathcal{F}\boldsymbol{J}_6$
电荷	相域（手性）	$Q = \text{sgn}(\text{Link})$
自旋	相域（扭转）	$\boldsymbol{S} = \frac{\hbar}{2\pi} \oint \boldsymbol{\tau}, ds$
信息	相域（排列）	$S = k_B \ln \Omega(\Phi)$
意识	相域（网络相干）	$\Phi_{\text{consciousness}}$
暗物质	相域（大尺度尾迹）	$\rho_{\text{DM}} \propto 1/r^2$
暗能量	相域（边界漂移）	$\rho_{\Lambda} = \text{geometric residual}$

English

Observed Quantity	Projection Domain	Mathematical Formulation
Force	3-D flat space	$\boldsymbol{F} = -\nabla_{3\text{D}}\boldsymbol{J}_6$
Mass	3-D flat space (closed curvature)	$m = \frac{\hbar}{R_5 c} \cdot \frac{\text{Link}}{\chi}$
Momentum	3-D flat space (first-order moment)	$\boldsymbol{p} = \int \boldsymbol{J}_6 \cdot d\boldsymbol{r}$

Energy	Time domain	$E = \partial J_6 / \partial t$
Frequency / Spectrum	Frequency domain	$\omega = \mathcal{F}J_6$
Electric charge	Phase domain (chirality)	$Q = \text{sgn}(\text{Link})$
Spin	Phase domain (torsion)	$S = \frac{\hbar}{2\pi} \oint \tau, ds$
Information	Phase domain (configuration entropy)	$S = k_B \ln \Omega(\Phi)$
Consciousness	Phase domain (network coherence)	$\Phi_{\text{consciousness}}$
Dark matter	Phase domain (large-scale wake)	$\rho_{\text{DM}} \propto 1/r^2$
Dark energy	Phase domain (boundary drift)	$\rho_{\Lambda} = \text{geometric residual}$

五、人类感知的“时间” = 三重投影的混合

5. Human-Perceived "Time" = Superposition of Three Projections

中文

人类把时域、频域、相域的混合感受称为“时间”：

时间感	来源投影	物理量
流逝感	时域投影	能量变化率
节律感	频域投影	频谱峰值
秩序感	相域投影	拓扑序变化

这就是为什么时间如此难以理解 —— 因为它是三个独立物理域的同时感知，而不是一个独立维度。

English

Human intuitive time arises from superimposed signals from three separate projection domains:

Perceptual Feature	Source Projection	Physical Origin
Sense of flow	Time-domain projection	Rate of energy change
Sense of rhythm	Frequency-domain projection	Spectral peak distribution
Sense of order	Phase-domain projection	Evolution of topological order

The difficulty in defining time in fundamental physics comes from this composite nature; human time is not one single independent dimension, but simultaneous perception across three distinct mathematical domains.

六、结论 / Conclusion

中文

宇宙只有一个物理量：角动量。

所有人类感知的物理量，都是角动量在四个投影面上的影子。

- 三维平直空间的投影 → 力、质量、动量
- 时域的投影 → 能量
- 频域的投影 → 谱、量子态
- 相域的投影 → 电荷、自旋、信息、意识、暗物质、暗能量

物理学的历史使命，就是把“力”“质量”“能量”“电荷”“自旋”“意识”“暗物质”这些看似独立的概念，还原为同一个几何结构的投影。ANG 完成了这个还原。

物理学的终点 = 人类不再需要“力”“质量”“时间”“意识”这些词——只需要“角动量”和“投影”。

English

There is only one fundamental physical quantity in the universe: angular momentum.

All quantities observed by humans are shadows of angular momentum projected onto four sub-spaces.

- Projection onto 3-D flat space → force, mass, momentum
- Projection onto the time domain → energy
- Projection onto the frequency domain → spectra, quantum states

- Projection onto the phase domain → charge, spin, information, consciousness, dark matter, dark energy

The historical mission of physics is to reduce seemingly independent concepts such as force, mass, energy, charge, spin, consciousness and dark matter to projections from one unified geometric structure. The ANG framework achieves this reduction.

□("The end point of physics: humanity may discard separate terms for force, mass, time, consciousness)
(, and describe nature only with angular momentum + projection.")

参考文献 / References

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[2] Song, C. (2026). UVMM-ANG v5.7: Cross-Disciplinary AI Knowledge Migration Package.